

Topic: Relational Calculus — Self-Learning

Q. What is Relational Calculus? Differentiate between Tuple Relational Calculus (TRC) and Domain Relational Calculus (DRC), and compare Relational Calculus with Relational Algebra.

📌 **Definition to Remember**

Relational Calculus is a **non-procedural (declarative)** query language where the user specifies *what* data is needed without specifying *how* to retrieve it. It is based on **mathematical predicate logic** and comes in two forms: **Tuple Relational Calculus (TRC)** and **Domain Relational Calculus (DRC)**.

Relational Calculus (Self-Learning)

A. Tuple Relational Calculus (TRC)

- Variables range over **entire tuples (rows)**.
- **Syntax:** $\{ T \mid P(T) \}$
- T = resulting tuple variable
- $P(T)$ = predicate (condition) that the tuple must satisfy
- **Read as:** "Find all tuples T such that $P(T)$ is true."
- **Example:** Find names of employees in the IT department:
 $\{ T.Name \mid Employee(T) \text{ AND } T.Department = 'IT' \}$

B. Domain Relational Calculus (DRC)

- Variables range over **individual attribute values (domains)** — not whole tuples.
- **Syntax:** $\{ \mid P(x_1, x_2, \dots, x_n) \}$
- x_i = domain variables (individual column values)
- P = condition on those variables
- **Example:** Find names of IT employees:
 $\{ \mid \exists ID, Salary (\in Employee) \}$

2. Relational Algebra vs Relational Calculus

Feature	Relational Algebra	Relational Calculus
Nature	Procedural	Non-Procedural (Declarative)
Specifies	What AND How to retrieve	What to retrieve only
Foundation	Mathematical Set Theory	Mathematical Predicate Logic
Variables	Operators on Relations	Tuple variables (TRC) or Domain variables (DRC)
Execution Order	Order of operations matters	DBMS optimizer determines execution
SQL Equivalent	Maps to query execution plans	Maps to SQL query syntax

★ Must-Write Points (for 10 marks)

1. Relational Calculus is **non-procedural** — specifies *what* to retrieve, not *how*.
2. Based on **mathematical predicate logic** (unlike Algebra, which is based on Set Theory).
3. Two types: **TRC** (variables = whole tuples) and **DRC** (variables = individual attribute values).
4. TRC syntax: $\{T \mid P(T)\}$ — find all tuples T where predicate $P(T)$ is true.
5. DRC syntax: $\{\mid P(x_1, x_2, \dots)\}$ — find specific domain values satisfying P .
6. Relational Algebra is procedural; Relational Calculus is declarative (like SQL).
7. Both are **relationally complete** — anything expressible in one can be expressed in the other.

⚡ Quick Recall

Relational Calculus → Declarative (What, not How) → Predicate Logic → TRC (tuple variables) → DRC (domain/attribute variables) → Foundation of SQL